

Caring for Patients With Advanced Chronic Kidney Disease: Dietary Options and Conservative Care Instead of Maintenance Dialysis



Kamyar Kalantar-Zadeh, MD, MPH, PhD,* Vincenzo Bellizzi, MD, PhD,† Giorgia B. Piccoli, MD,‡ Yunying Shi, MD, MHA,§ Soo Kun Lim, MD,¶ Sumira Riaz, DPsych,** Rocio Urbina Arronte, MD,†† Wai Pooi Lau, BSc,‡‡ and Denis Fouque, MD, PhD§§

An expert advisory board discussed the prevention and treatment of chronic kidney disease (CKD), with a focus on dietary options. This is timely, given the uptake of value based models for kidney care in the United States. Timing of dialysis start is influenced by patients' clinical status and complex patient–clinician interactions. Patients value personal freedom and quality of life and may want to delay dialysis, whilst physicians are sometimes more concerned with clinical outcomes. Kidney-preserving therapy can prolong the dialysis-free period and preserve residual kidney function, thus patients are asked to adjust their lifestyle and diet, to follow a low- or very low-protein diet, with or without ketoacid analogues. Multi-modal approaches include pharmacotherapies, management of symptoms, and a gradual, individualized dialysis transition. Patient empowerment is vital, including CKD education and involvement in decision making. These ideas may help patients, their families, and clinical teams to improve the management of CKD.

Keywords: chronic kidney disease; nutrition; low-protein diet; ketoacid analogue supplements; dialysis; value-based models

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Introduction

THE GLOBAL BURDEN of chronic kidney disease (CKD) has increased substantially over the last 3 decades. About 1.2 million people were estimated as dying

worldwide from CKD in 2017, representing an increase of 42% since 1990.¹ It has been predicted that CKD will become the fifth leading cause of mortality by 2040.² CKD occurs most commonly in older people, and is among

*Division of Nephrology, Hypertension and Kidney Transplantation, University of California, Orange County, California.

†Nephrology and Dialysis Division, Department of Medical Sciences, Hospital Sant'Anna e San Sebastiano, Caserta, Italy.

‡Service de Néphrologie, Department of Medicine, Centre Hospitalier Le Mans, Le Mans, France.

§Department of Nephrology, West China Hospital of Sichuan University, Chengdu, Sichuan, China.

¶Department of Medicine, Faculty of Medicine, University of Malaya, Kuala Lumpur, Malaysia.

**Royal National Orthopaedic Hospital, London, UK.

††Fresenius Kabi Nutritional Care Center, Mexico City, Mexico.

‡‡Department of Dietetics, University Malaya Medical Centre, Kuala Lumpur, Malaysia.

§§Department of Nephrology, Centre Hospitalier Lyon Sud, Université Claude Bernard Lyon 1, Cardiometabolism and Nutrition (CarMeN), Lyon, France.

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Address correspondence to Denis Fouque, MD, PhD, Department of Nephrology, Centre Hospitalier Lyon Sud, Université Claude Bernard Lyon 1, Cardiometabolism and Nutrition (CarMeN), Lyon, France. E-mail: denis.fouque@univ-lyon1.fr

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Box 1**The rationale for low-protein diets and supplemented very low-protein diets in preservative management of CKD.**^{14,22–36}

High dietary protein intake (>1.5 g/kg ideal body weight [bw]/day)²⁶ can cause glomerular hyperfiltration and inflammatory gene expression, which are risk factors for kidney disease.^{14,22–24} A high-protein diet dilates the glomerular afferent arteriole, leading to increased intraglomerular pressure, resulting in a higher glomerular filtration rate over a short period.^{14,25} Ultimately, this glomerular hyperfiltration stimulates mesangial-cell signaling, leading to increased transforming growth factor beta release and thereafter to progressive fibrosis and renal damage.¹⁴ In contrast, a low-protein diet has the reverse effect—constricting the afferent arterioles and lowering intraglomerular pressure.¹⁴ Most controlled trials and meta-analyses have demonstrated beneficial effects of restricting protein intake, as well as benefits for replacing red or processed meats with plant or dairy sources.^{14,27,28} One exception was the Modification of Diet in Renal Disease study, that showed minimal deceleration in the progression of kidney disease with a low-protein diet,²⁹ though secondary analyses tended to show a beneficial effect of protein restriction.³⁰

The rationale for very low-protein diets includes evidence that the lower the protein intake the greater the benefits, given the lack of threshold effects between reductions in dietary protein intake and lowered CKD risk.³¹ A very low-protein diet typically involves a reduction in protein intake to 0.3 to 0.4 g protein/kg bw/day with the addition of KA supplements (in settings where KAs are available) to provide adequate total protein/amino acids and essential amino acids to meet nutritional needs.³² Moreover, KAs provide essential amino acid precursors with reduced nitrogen levels, and so may generate fewer toxic metabolic products than protein from low-protein diets.³³ Nevertheless, patient adherence to very low-protein diets can be challenging, as shown in longer-term studies,³⁴ underscoring the importance of good patient support.^{35,36}

the 10 main causes of disability-adjusted life years in those aged 50 years or older.³ Comorbid diabetes, hypertension, and cardiovascular disease, as well as poor nutrition, are the most important predictors of poor outcomes.⁴ The global prevalence of CKD is about 9%, and while age-standardized prevalence has remained constant over recent decades, the all-age prevalence of CKD increased by 29% between 1990 and 2017.¹ This may explain the rise in prevalence in Western countries in-line with aging population demographics.⁵ However, the sharp rise of CKD prevalence in emerging countries may be more closely linked to increased rates of diabetes and obesity rather than increased life expectancy.⁵ The global health burden of CKD also falls disproportionately on the poor and populations in low- and medium-income countries.^{1,6}

Most people with CKD will inevitably progress to kidney failure, known as ‘end-stage renal disease’ (ESRD) which is incompatible with life without renal replacement therapy (RRT) in the form of either maintenance dialysis treatment or kidney transplantation.⁷ Patients receiving maintenance dialysis often have very high rates of cardiovascular morbidity and mortality, with an overall 5-year survival of less than 50%.^{7,8} One of the choices faced by patients with CKD and their clinicians is when to start dialysis. An early start to dialysis according to estimated glomerular filtration rate (e.g., 10–15 mL/minute) does not necessarily improve patient outcomes, and can lead to higher costs and lower quality of life.^{9–11} In settings where RRT is readily available, such as the United States (US) and Western Europe, it accounts for a disproportionate share of healthcare budgets (2%–5% of overall healthcare costs for

about 0.1% of the overall population).⁵ Different types of RRT differ in associated costs and benefits: transplantation has the lowest cost and the highest quality of life, while in-hospital hemodialysis accounts for the highest cost and provides the lowest quality of life.⁹ The choice of dialysis or transplantation is, however, also determined by clinical conditions and by the availability of organs for transplantation. In addition, dialysis is either not available or not affordable for around two-thirds of the world’s population, and even in high-income countries limited resources (among other factors) restrict transplants or dialysis in the very old or patients with numerous/severe comorbid disorders.¹² Therefore, it is important to also consider options beyond the simple dichotomy of RRT or conservative treatment.

CKD is largely preventable and treatable, and often the use of interventions can avoid or postpone adverse outcomes.^{1,5,13} Thus, interventions to preserve kidney function are increasingly advocated for reasons such as improved healthcare effectiveness and better patient quality of life. An important goal of this approach is to prolong and preserve kidney function, and so defer starting dialysis. In order to achieve this goal patients are asked to change their diet, in particular they are asked to reduce their protein intake (e.g., to a low- or very low-protein diet),¹⁴ to eat less salt, and perform regular physical exercise to improve body and organ/system fitness.^{15,16} Changing diet or building physical exercises into a daily schedule generally means changing lifestyle, and this poses a number of challenges for the patient concerned, his/her caregivers, and healthcare professionals (HCPs). This approach involves increasing awareness of preventive measures in patients, professionals,

and policy makers, and ensuring that patient management includes education, empowerment and motivation.¹⁷ Furthermore, delaying or deferring dialysis has become an even higher priority given the very recent adoption of value-based models in the US, which offer rewards and financial incentives to delay the start of dialysis in patients with late-stage CKD.

Patient choice is frequently given a low priority when deciding to start dialysis, which frequently tends to be precipitated by clinical events involving acute illness or medical procedures.¹⁸ In these settings, physician recommendations to start dialysis sometimes conflict with patient priorities.¹⁸ Furthermore, in developed countries some patients, whose lives might not be extended by dialysis, may be 'over-treated' (e.g., frail, older patients, and those with multiple comorbid conditions).¹⁹ Thus, there is increased movement away from dialysis toward alternative interventions such as dietary changes. However, while this may lead to societal benefits, it is crucial to respect and protect patient choices to delay, avoid, or withdraw (or indeed continue with) dialysis.^{17,19,20}

An international group consisting of nephrologists, dietitians and a psychologist met for a virtual advisory board (3 May and 7 May, 2021) to exchange experience on the aforementioned subjects of prevention and treatment of CKD, as well as patient choices. In addition, the group considered potential practical and educational tools to support patient adherence to nutritional/behavioral changes. This narrative review, authored by the virtual advisory board members, summarizes the key aspects of the ideas and outcomes presented and discussed at this meeting to help inform patients, their families, and clinical teams involved in the management of CKD.

Issues to Overcome—Medical Aspects

In recent years there has been a paradigm shift in the treatment of CKD away from the dichotomy of dialysis versus no dialysis (conservative or supportive treatments) to include a range of other intermediate options and the use of more integrated approaches. These can be grouped together as preservative management of CKD, defined as a type of life-sustaining conservative management with the primary goal of slowing CKD progression and preserving kidney function for as long as possible, ideally avoiding dialysis altogether.¹⁹ This approach strives to improve survival rates, including the improvement of cardiovascular health and patients' health-related quality of life, with effective treatment of renal and non-renal comorbidities and associated symptoms.¹⁹ Preservation of kidney function can be achieved through non-pharmacological strategies (i.e., dietary and lifestyle adjustments) and/or pharmacological interventions targeting CKD or kidney disease.¹⁹ While non-pharmacological strategies (principally dietary and lifestyle interventions) are the main topic of the current article, it is expected that these strategies can have a syner-

gistic effect on concurrent pharmacotherapies such as angiotensin-pathway modulators and do not interfere with other aspects of patient care.^{19,21}

The role of low-protein diets and supplemented very low-protein diets in preservative management of CKD are summarized in Box 1.^{14,22–36} High-protein diets have gained in popularity in recent years for people seeking weight loss or to ameliorate type-2 diabetes, and in general, high-protein intake can be defined as greater than 1.5 g/kg ideal body weight (bw)/day or dietary proportion of >25% of the daily energy intake, though definitions are not always clear.²⁶ However, high-protein diets can cause intraglomerular hypertension, potentially leading to renal hyperfiltration, glomerular injury, and proteinuria, and in time may adversely affect kidney health in populations with or at-risk for CKD.^{7,26} Moreover, dietary-protein quality as well as quantity may play a role: several observational studies have shown that animal protein, particularly processed red meats, is associated with an increased risk or progression of CKD compared with protein from plants.^{26,37,38} While red and/or processed meat were associated with increased risk of albuminuria, rapid estimated glomerular filtration rate decline, or both,^{37,39} reducing the intake of animal protein and increasing the dietary intake of nuts, legumes and low-fat dairy products was associated with a protective effect against the development of CKD.^{37,38}

The recommended dietary allowance for protein intake in a normal healthy adult is 0.8 g/kg bw/day, estimated from mean protein requirements of non-pregnant, non-lactating adults (0.6 g/kg bw/day) including an additional 33% (0.2 g/kg bw/day) safety margin.³³ Patients with stable (non-nephrotic, non-inflamed, non-catabolic) non-dialysis dependent CKD have similar protein requirements (i.e., ~0.6 g/kg bw/day).^{32,33} As such, provided there is sufficient energy intake (i.e., at least 25 or 30 kcal/kg bw/day), dietary protein intake can be decreased safely to 0.55 to 0.6 g/kg bw/day (low-protein diet), and an additional reduction in protein intake to 0.3 to 0.4 g protein/kg bw/day (very low-protein diet) may be achieved with the addition of ketoacid analogue (KA) supplements (in settings where KAs are available) to provide sufficient essential amino acids normally supplied by animal proteins, which are typically scarce in these very low-protein/vegan-like diets.³²

The potential role of low-protein diets in patients with CKD has been hampered by fears that restriction of dietary protein may lead to protein-energy wasting (PEW).⁴⁰ However, most evidence indicates that a carefully controlled low-protein diet or supplemented very low-protein diet, combined with an adequate intake of energy, is more likely to prevent rather than promote PEW.³³ Moreover, studies of protein restriction in patients with CKD without other serious illnesses have shown good safety profiles and low malnutrition rates, even with very low-protein diets,^{28,41,42} or low-protein diets in patients with diabetes mellitus.^{43,44} Indeed, malnutrition is not

generally observed for patients on a very low-protein diet supplemented with KAs and with adequate energy intake, though to ensure proper nitrogen metabolism and avoid PEW in CKD requires strict adherence to the dietary energy prescription during protein restrictions.⁴⁵ Patients at increased risk for PEW may also receive particular benefit from supplements (e.g., KAs).¹⁴ Potential benefits of supplementation with KAs include the maintenance of protein–energy status without increasing nitrogen waste products; reducing phosphorous, nitrogen and acid load, decreasing protein degradation and enhancing protein synthesis.^{40,46,47}

In settings where dialysis is readily available it has often become the treatment of choice for patients with ESRD, regardless of their age, fragility or comorbidity burden; however, in the presence of multiple comorbidity, dialysis may not increase life expectancy, and can impair quality of life, resulting in over-treatment.^{18,20} As a consequence, there is a growing trend toward expanding patients' choices past the perceived dichotomy of dialysis versus no dialysis. Beyond the conservative and preservative management of CKD, which includes the use of low-protein diets and lifestyle modifications, as discussed above, pharmacotherapies, and management of symptoms such as pain and fatigue, there are other options such as a gradual transition to dialysis or supportive care. A gradual transition may be starting once- to twice-weekly hemodialysis, or less-than-daily peritoneal dialysis at home, each of which may preserve residual kidney function longer than conventional dialysis.²⁰ Moreover, a low-protein intake may be used as part of the nutritional management during a gradual transition to dialysis therapy, as it may help delay not only the start of dialysis but also the need for more frequent hemodialysis.⁴⁸ Thus, once-weekly hemodialysis, though often considered as just a bridge to more frequent (e.g., thrice-weekly) dialysis, can, when combined with a low-protein diet, delay this progression.⁴⁸ However, when considering incremental dialysis, it is also important to tailor dialysis frequency and dose according to patients' residual kidney function, with such an approach being more optimal for those starting treatment with considerable native kidney function.^{49–51} Furthermore, an individualized approach may be most appropriate, taking into account factors such as life expectancy, comorbidity, nutritional status, as well as residual kidney function.⁵²

A range of alternative options may be more appropriate in patients close to the end of their lives, such as the expanded use of supportive care, and/or gradual decrease in the frequency and intensity of dialysis. Symptom management and patient comfort may become the most important goals in these cases rather than aggressive clearance or rigorous fluid removal. Furthermore, the dialysis setting could encompass hospice care, and home dialysis could be offered at nursing homes or in skilled nursing facilities.²⁰

However, patient choice should be paramount, and respected, regardless of whether they choose to receive comfort care, withdraw from dialysis, have less frequent or shorter dialysis, or continue with their dialysis treatment.

Issues to Overcome—Patient-Related Aspects

There is a growing realization that the priorities of health-care staff involved in the treatment of CKD are to naturally focus primarily on patients' clinical outcomes and correction of biochemical alterations, and that these may be very different to the priorities of their patients. A survey that investigated the reasons behind choosing continued conservative care or starting dialysis in patients with CKD stage 3 to 5 showed that some patients preferred conservative care regardless of survival benefits afforded by dialysis, while other patients would not consider dialysis if it involved too many hospital visits or travel restrictions.⁵³ Thus, freedom can sometimes be more important to patients than clinical outcomes: in fact, patients were willing to forgo an average of 7 months of life expectancy to reduce the number of hospital visits for dialysis and forgo 15 months of life expectancy to increase their ability to travel.⁵³ The potential differences in priorities between patients with CKD and healthcare staff makes consideration of the patient perspective even more important. The timing of dialysis therapy initiation, in particular, can be shaped by interactions between patients and clinicians; often patients may want to delay dialysis while physicians are more concerned with safety and survival, seeing delays in starting dialysis as potentially hazardous.¹⁸ Meanwhile, patients often have different concerns, emphasizing quality of life and autonomy.¹⁸ As such, it is vital that enhanced efforts are made to improve patient–physician communication, in particular to clarify patients' healthcare values, goals, and preferences, and to better align treatment recommendations with patients' priorities.

Traditionally, medical care for patients with CKD has been viewed as rather paternalistic, with clinicians typically making treatment decisions independently of patient preference, but shared decision making and patient empowerment are key aspects of CKD management in the 21st century. Shared decision making requires that medical choices are made collaboratively, typically in accordance with evidence provided by the clinician and the values and preferences of the patient.⁵⁴ Patient empowerment can be defined as “a process through which people gain greater control over decisions or actions affecting their health”.⁵⁵ This translates into patients having sufficient knowledge to engage with clinicians in shared decision making, and to have the skills and support needed for self-management of CKD.¹⁷

Patient empowerment can be stimulated by providing CKD-related patient information and education as well as whole-heartedly engaging with patients to involve them fully in the decision-making processes concerning their

Table 1. Suggested Strategies for “Living Well With CKD” Using a Strengths-Based Approach¹⁷

Strengths-based approach	Suggested Strategies
Build resilience	– Identify or provide strategies and resources to manage stress and functioning when encountering challenges, adversity and trauma (e.g., commencement of dialysis)
Harness social connections	– Facilitate connections with other patients to learn coping strategies and for support – Support family members/caregivers
Build awareness and knowledge	– Provide education (including practical advice) on diet and lifestyle modifications – Understand, identify, and address the potential impacts of CKD (e.g., cognitive function) – Encourage patients to ask questions – Encourage the use of knowledge to empower and prepare for the future
Facilitate access to support	– Refer to allied health care professionals (e.g., dietitian, social worker, mental health professionals, occupation therapists) – Provide support that enables the patient to participate in important life activities (e.g., work)
Establish confidence and control in self-management	– Support informed and shared decision making (including dialysis, kidney transplantation, conservative or non-dialytic care) – Encourage patients to learn to “get in tune” with what works well for them and to voice any concerns, and work together to develop better management strategies to enable patients to feel better – Provide strategies to prevent or manage complications (e.g., infection) – Support open communication regarding goals, concerns, and priorities

CKD, chronic kidney disease.

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care.¹⁷ Effective empowerment strives to minimize CKD-related symptoms to help enhance patient satisfaction, health-related quality of life, and life participation.¹⁷ CKD interventions targeting patient knowledge and empowerment have been shown to help improve health-related quality of life, medication adherence, and blood pressure.^{56–59} Moreover, a survey of patients with CKD undergoing dialysis indicated that their sense of empowerment was strengthened by understanding the clinical rationale underpinning lifestyle restrictions, having ongoing practical training, and receiving the support of their family in managing their treatment regimens.⁵⁶

Multi-disciplinary healthcare teams have a vital role to play to align patients and healthcare teams, improve outcomes, and also fulfill patients’ priorities such as maintaining their quality of life and independence. The involvement of a multi-disciplinary team can be particularly crucial for patients with CKD, as this disease presents with multiple complications that worsen with progression, and so management requires a careful and integrated approach across all comorbidities and complications, up to and including renal transplantation.⁶⁰ Multi-disciplinary care can have considerable benefits. For example, a multi-disciplinary approach improved the outcomes of children with CKD, particularly for the management of anemia, bone mineral metabolism, nutrition, and slowing renal disease progression.⁶¹ Other studies of multidisciplinary CKD clinics have shown improvements in fistula rates, hospitalization, and mortality rates, as well as showing this to be a cost-effective intervention.⁶²

A core multi-disciplinary team could typically consist of a nephrologist, renal nurses, dietitians, pharmacists, physiotherapists/exercise physiologists and social workers, and sometimes a psychologist (or other healthcare workers trained in psychosocial support of patients with CKD, such as social workers, medical or psychiatric nurses, or physicians).^{62,63} Each team member has specific skill sets dedicated to preventing CKD progression, managing complications of CKD, or managing the transition to dialysis, transplant or conservative therapy. The role of a psychologist, or other team members offering psychosocial support, is of increasing importance within CKD multidisciplinary teams, possibly because this disease can have profound negative biopsychosocial effects. For example, patients undergoing dialysis commonly experience anger, depression, and feelings of hopelessness.⁶⁴ Moreover, increased negative perception of the effects of CKD associate independently with increased mortality rates in patients undergoing dialysis.⁶⁵ Psychosocial interventions for patients with CKD may reduce depressive symptoms and anxiety, and improve health-related quality of life.⁶³ Overall, beyond specific psychological interventions, strategies to build coping and empowerment skills may be most useful, such as in building patient awareness and knowledge, supporting patients/caregivers to self-manage, developing resilience and self-confidence in relation to lifestyle choices, and setting realistic, achievable goals.¹⁷ This so-called ‘strength-based approach’ acknowledges that each patient has the strengths and abilities to overcome their problems and challenges (Table 1).¹⁷

Experience and Approaches from Different Countries

A national survey in France has assessed the association between different stages of CKD and the physical and mental health domains of health-related quality-of-life scores.⁶⁶ In essence, health-related quality-of-life physical scores were significantly lower in patients within all CKD subgroups than in the general population, with the most marked impairment in patients receiving dialysis, whereas for mental scores only dialysis patients represented a clinically important difference to the general population.⁶⁶ Thus, quality of life is particularly impaired during dialysis, and so one of the best ways to improve quality of life in patients with CKD is to delay ESRD and dialysis by all possible interventions, including low-protein diets. One worry has been that low-protein diets may reduce patients' quality of life. However, recent studies have shown that low-protein diets were not associated with impairment of quality of life in patients with advanced CKD and were compatible with high dietary satisfaction.^{67,68}

In Italy, nephrologists have a well-established tradition in the implementation of low-protein diets for the treatment of patients with CKD, following the objectives of alleviating uremic symptoms, improving nutritional status, and also slowing the progression of CKD or delaying the start of dialysis.^{41,69} Another aim is to improve the nutritional aspects in patients with CKD by providing adequate energy intake, including protein of high biological value, where necessary supplemented with essential amino acids or KAs.⁶⁹ The flexible and innovative Italian approach to the low-protein diet in patients with CKD also includes the improvement of patient adherence, which is critical for the implementation of a low-protein diet program.⁶⁹⁻⁷¹ This may require good knowledge and understanding on behalf of the patient, access to dietitians, individual dietary programs, intensive educational programs, and counseling, though dietitians are not commonly available in clinical practice in Italy.⁷² This limitation may be overcome by using simplified approaches to a conventional low-protein diet, which may be practiced when dietitians are not available.^{69,70,73}

The role of specialist dietitians is of increasing importance within nephrology (e.g., so-called 'renal dietitians'), given the renewed interest in diet-related interventions in this field. However, in many countries such as Malaysia there is a shortage of dedicated dietitians involved in the care of patients with CKD, and so nutritional management is primarily carried out by physicians and nurses.⁷⁴ This can hinder patient-centered approaches such as personalized nutritional interventions as well as the implementation of standardized nutritional guidelines.⁷⁴ However, when nephrologists and dietitians work together, dietitians can play an active role in preventing or slowing down the progression to stage 5 CKD. For example, clinical staff can explain general principles of dietary restriction to the pa-

tient (reduced protein, sodium and animal protein intake, particularly 'red meat'), before referring them to the renal dietitian for practical treatment advice. In Malaysia, eating out is common as are high-protein diets, similar to Western countries. It is a common misconception that Asian countries have low protein diets: in Malaysia, for example, the mean protein intake is far higher than recommended levels, particularly among men (i.e., 73 g/day vs recommended levels of 51 to 62 g/day), while mean fruit and vegetable intake is lower than is recommended.⁷⁵ Approaches such as gradual, step-wise reductions in dietary protein have been successful, following the principle that even a reduction of 0.2 g protein/kg bw/day can ameliorate glomerular filtration rate decline by 29% and halve the risk of renal failure or death.³⁰ Protein restriction plus the substitution of protein from plant sources rather than meats can also reduce intake of phosphorous, purine, and fat intake, which may be beneficial for renal patients. Another potential option is low-protein food versions of bread, pasta and cookies, that are well distributed and used in other countries, which could replace their regular parent products and allow a reduction of protein intake by 0.2 g/kg bw/day without requiring much dietary knowledge or effort. Other approaches include setting goals according to each patient's readiness to change, and may be most effective by addressing one opportunity at a time rather than a broad long-term diet planning strategy.⁷⁶

In Mexico, CKD care is suboptimal, with limited access to medical treatments for CKD owing to a lack of a universal healthcare system and a high proportion of the population lacking health insurance.⁷⁷ About half of the population does not have access to social security benefits and cannot afford private healthcare services, and so has limited or non-existent access to dialysis, and are highly unlikely to get pre-dialysis nephrology care.⁷⁸ One intervention that has proved successful in Mexico is the founding of nutritional care centers 'Centros de Atención Nutricional de Fresenius Kabi México' (CEAN) centers. These were created with the objectives of prolonging the predialysis stage, giving nutritional assistance and education to achieve an adequate nutritional status, better metabolic control, and good adherence to low-protein diets in patients with CKD taking KAs. The first CEAN center started in 2006, and currently there are currently 8 centers (including one virtual center) in Mexico with a total of 15 nutritionists treating around 3,500 patients with CKD that are referred by their nephrologist or internist to take a low-protein diet supplemented with KAs. The approach taken is to perform a detailed nutritional assessment in which a nutritional diagnosis is made, leading to individualized diet plans supported by educational materials and followed by ongoing nutritional monitoring.

CKD is highly prevalent in China, and a growing problem: the number of dialysis patients in China has doubled

Table 2. Main Issues and Concerns Faced by Patients With CKD and Their Carers, and Some Possible Solutions

Issues and/or Concerns	Suggested Strategies
Restricted, insufficient or lacking access to the multi-disciplinary healthcare team to address: – clinical aspects of CKD, e.g., to determine shared priorities for patient management – psychological aspects of CKD, such as emotional distress (in early as well as late disease course) or lack of patient motivation – foster patient motivation and empowerment to promote adherence over the longer term.	Allocate more human and financial resources to CKD, particularly to aspects related to the preservative management of CKD such as: – increased general consultancy time by physicians – improved access to psychological support (either directly with a psychologist, or indirectly via psychological training of other staff) – one-to-one consultations and dietary plan discussion with a dietitian. Involvement of the multi-disciplinary team to set achievable goals, monitor, and offer support. Collaborate with patient societies and/or industry for programs that improve patient support and make more widespread use of remote monitoring and apps (e.g., dedicated tablets may help older patients).
Little understanding, awareness, and knowledge of disease trajectory.	Develop (or improve) educational programs, so that they: – supply information in an incremental manner (rather than all at once) – convey positive rather than negative information (e.g., diet can have beneficial effect; detail what you can rather than what you cannot eat) – address misinformation/confusing information. Use modern tools such as apps and social media. Dedicated tablets may help older patients. Ensure that the multi-disciplinary team is aligned and conveys similar messages (e.g., about the choices beyond the dichotomy of dialysis or no dialysis). Educate carers and families of patients too.
Perceptions and fears related to CKD, such as: – difficulties understanding and balancing the therapeutic options including aspects of quality of life and longevity – negative effects of CKD on the life of the carer or partner – financial impact of loss of livelihood of the patient, and potentially their carer too.	To allow patient empowerment and shared decision making, outline the different therapeutic options including anticipated therapeutic effects and aspects of quality of life and longevity, for example: – low-protein diets or very low-protein diets plus KAs and healthy lifestyles can delay or prevent the need for dialysis, and that it is possible to have a good quality of life (and longevity) in many cases without dialysis – when dialysis is needed an individualized plan to start (incremental dialysis) is often the best option – numerous people live with dialysis for many years, and though dialysis can have a profound effect on quality of life it is manageable and a life-saving therapy. For financial concerns related to CKD, initiate programs and practical support measures (e.g., social worker help and advice) to help with these aspects.
Perceptions and fears related to the new dietary plan, such as: – meal plans do not fit with patients' culture, lifestyle, or lack of perceived choice, or that healthy food is tasteless – fear of potassium burden in primarily plant-based diets	At initial consultations: – receive nutrition counseling from a dietitian for a personalized diet prescription and meal plan – adapt recipes based on cultural norms or lifestyle, so that patients can meet their dietary goals – underscore the beneficial effects of a plant-based diet (values of low protein and high fiber content), and that diet and lifestyle are 'powerful medicines.' Regular follow-up by dietitians to monitor adherence to meal plans and to foster patient motivation.
Symptom management issues: – physical pain and emotional distress – comorbid conditions such as hypertension, heart disease, cancer, or depression	Greater awareness of these issues as barriers to better healthcare of patients with CKD.

CKD, chronic kidney disease; KA, ketoacid analogue.

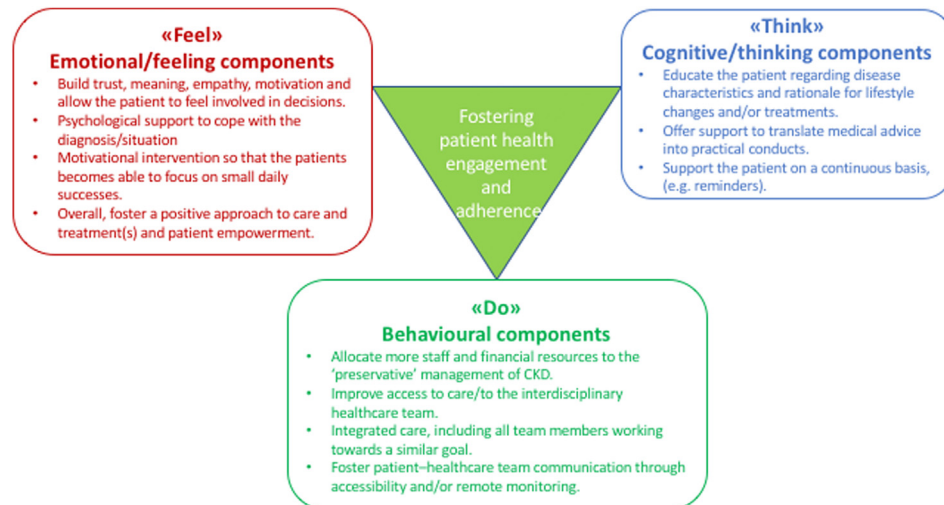


Figure 1. Proposed actions for healthcare teams that may help to foster patient health engagement and adherence. CKD, chronic kidney disease.

between 2012 and 2017.⁷⁹ According to the Chinese National Renal Data System,⁸⁰ there were a total of 524,467 prevalent patients on hemodialysis and 86,344 on peritoneal dialysis by 2017.⁷⁹ The rapid rise in numbers of patients receiving dialysis in China is likely to be because of several factors. These include an increased prevalence of CKD risk factors such as diabetes mellitus and hypertension,^{81,82} plus increased coverage for dialysis via social insurance in recent years, in conjunction with dialysis being offered increasingly to patients.⁸³ Other factors include increasing health literacy in developed areas of China, together with the largest rural-to-urban migration in human history leading to improved access to health services and higher incomes.^{83,84}

Given these increasing demands on healthcare services associated with CKD and limited numbers of nephrologists, specialist CKD nurses, and dietitians in China, such HCPs may often only interact with each patient for a few hours a year. As such, in China it is particularly important to develop patients' self-management by improving their diet and lifestyle, as well as compliance to treatment, and providing education remotely. New ways to educate and motivate patients are being used. These include CKD management software, both in the clinics and linked to patient apps, extending management from hospital to home, encompassing real-time control and information exchange. Other measures include individualized patient education materials, telephone follow-up calls, on-line patient lectures and information videos, and 'chat apps' for patient groups to provide peer support. A recent example is 'My Kidneys & Me,' an online self-management and lifestyle education program, to enable accessible, tailored, and interactive information and support for patients with CKD.^{85,86} By using these methods, lifestyle and diet information can be provided, which is particularly useful to address lack of information,

prejudice concerning low-protein diets, and other issues such as problems with recipes and diet tolerability.

The World Kidney Day is a joint initiative of the International Society of Nephrology and the International Federation of Kidney Foundations to raise awareness of renal health and to reduce the frequency and impact of kidney disease and its associated health problems worldwide. 2021 was designated the year of 'Living Well with Kidney Disease' to improve CKD education and awareness of effective symptom management and patient empowerment, while continuing to emphasize measures to prevent kidney disease and its progression.^{17,87} This campaign advocates prevention of CKD progression by lifestyle and diet modifications, such as intake of low-protein primarily plant-based foods, and is applicable to both developed and low- and medium-income countries.¹⁷ Moreover, the recently introduced value-based model for Kidney Care by the US government endorses such an approach.^{88,89}

Future Directions

A key part of the discussion summarized in this review paper was to address the main issues and concerns faced by patients with CKD and their carers and also to outline some practical solutions, where possible. The outcome of these discussions is summarized in Table 2. Some of the key points raised are the importance of the involvement of multidisciplinary teams, including a dietitian and psychologist. However, it was mentioned that dedicated psychologists are often not available, particularly for patients with CKD who are not undergoing dialysis. One solution to this is to use psychologists to provide educational programs and to educate the clinical team who can then administer some psychological support. Other key points identified were better education, both of patients and carers to counter misinformation and lack of information, as well

Table 3. The Continuum of Care Across CKD Stages^{32,44,87,90-92}

	CKD stage 1: at increased risk/initial kidney damage	CKD stage 2: mild loss of kidney function	CKD stages 3A–3 B: moderate-to-severe loss of kidney function	CKD stage 4: severe loss of kidney function	CKD stage 5: kidney failure	
eGFR (mL/min/1.73 m ²)	≥90	89-60	59-30	29-15	<15 (non-dialysis)	<15 (dialysis)
Prevalence (% of adult (USA) population)*	4.7%	3.3%	6.4%	0.4%	0.1%	
Prevalence (%) ACR ≥30 mg/g*	4.68%	3.33%	1.68%	0.26%	0.1%	
CKD prevention†	<i>Primary prevention:</i> – identify high-risk persons – reduce obesity – improve glycemic and BP control – encourage healthy diet and lifestyle, including reducing salt intake <i>Secondary prevention:</i> – goals are slowing CKD progression and delaying transition to dialysis – requires early CKD detection, better BP control, lower protein- & higher plant-based diet			<i>Tertiary prevention:</i> – goals are the management of uremic symptoms and comorbidities (e.g., anemia, acidosis, mineral-bone disorders, fluid overload, cardiovascular disease, protein-energy wasting) – requires uremia and fluid management, tailored cardiovascular and nutritional therapy, and pharmacotherapy		
Protein intake (g/kg BW/d)‡	~0.8—as per estimated normal optimal daily needs	<i>If metabolically stable, no dialysis:</i> 0.55-0.60 (LPD) or 0.28-0.43 + KA (to meet protein requirement of 0.55-0.60; VLPD) <i>In patients with diabetes, no dialysis:</i> 0.6-0.8. Although KA-supplemented LPD and VLPD are not mentioned explicitly in the 2020 KDOQI guidelines for patients with CKD and diabetes,‡ authors of this article report anecdotal and published data, as well as clinical experience, of using of KAs in this group. Additionally, in a recent meta-analysis of 11 studies¶ (not available at the time of compiling the KDOQI guidelines), renal outcomes (including GFR and 24-h urinary protein excretion), metabolic outcomes (including serum urea and blood glucose), clinical outcomes (including blood pressure and anemia), and nutritional outcomes (including serum albumin and body weight), were all in favor of using KAs in patients with CKD and diabetes, and these dietary regimens were safe and well tolerated, with no evidence of adverse effects on nutritional status.				1.0-1.2 if metabolically stable, but for patients with diabetes at risk of hyper- and/or hypoglycemia, higher levels of dietary protein intake may need to be considered to maintain glycemic control
Prevalence of metabolic complications:§						
– hyperparathyroidism	–	20%	30%-60%	75%-85%		
– anemia	–	10%	10%-20%	20%-40%		
– acidosis	–	<5%	5%-10%	20%-40%		
– hyperkalemia	–	<5%	5%-15%	20%-40%		
– hyperphosphatemia	–	<5%	<5%	5%-30%		
Frequent clinical symptoms (prevalence >50%)	Bone/joint pain, muscle cramps, feeling irritable, feeling sad, lack of energy, lethargy, difficulty sleeping, difficulty concentrating, trouble with memory, loss of appetite, heartburn.			Additional symptoms: swelling of arms or legs, difficulty breathing, difficulty keeping legs still, numbness/tingling in hands and feet, dry skin, itching.		

ACR, albumin-to-creatinine ratio; BP, blood pressure; BW, body weight; CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate; GFR, glomerular filtration rate; KA, ketoacid analogue; KDOQI, Kidney Disease Outcomes Quality Initiative; LPD, low-protein diet; VLPD, very low-protein diet; USA, United States of America.

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as educational programs for clinical staff. Proposed actions for healthcare teams that may help to foster patient health engagement and adherence are shown in Figure 1.

It is important to emphasize to patients and clinicians that there are choices beyond the dichotomy of dialysis or no dialysis, and rather that there should be a continuum of care across CKD stages (Table 3).^{32,44,87,90–92} Moreover, it is possible to have a good quality of life and longevity, even without dialysis, provided patients adopt diet and healthy lifestyle changes. Moreover, patients should be made aware that CKD diets can align with general population healthy eating guidelines, and such positive messaging may improve acceptance among those with CKD.⁹³

Where dialysis is required, it may be appropriate and beneficial to offer an individualized plan to start gradually (incremental dialysis), and this also gives the patient time to get used to this modality. In addition, fears of dialysis need to be addressed. Although dialysis can adversely affect patients' quality of life, it is also a life-saving therapy. In the future, the panel anticipates the importance of making far greater use of technology such as apps and dedicated tablets to allow for more efficient monitoring and communication between patients and clinical staff. Moreover, as awareness increases among patients and clinical staff that there are choices beyond the dichotomy of dialysis or no dialysis, low-protein diets and KAs will become used more widely, offering the opportunity to improve patients' quality of life by delaying or even avoiding the need for dialysis. We hope that the publication of this article will help to foster patient-physician exchanges concerning CKD management options, which may be particularly useful as patients and doctors tend to have different priorities. Whichever path is taken, patients require support from their HCP team to succeed. A series of field-tested tools can assist patient adherence to nutritional changes, and the combined efforts of patients and their supporting team are needed to implement these changes successfully.

Practical Application

Interventions to preserve kidney function are increasingly advocated for reasons such as improved healthcare effectiveness and better patient quality of life. The aim of these interventions is to defer or even avoid dialysis. Patients are asked to adjust their lifestyle and diet, to follow a low- or very low-protein diet, with or without ketoacid analogues. Such lifestyle modifications are associated with numerous hurdles, and so multi-disciplinary involvement, patient empowerment, and the use of technology are also key approaches to improve CKD care. This manuscript provides expert practical advice from the US, France, UK, Italy, China, Malaysia, and Mexico, supporting patients with CKD on their journey toward a healthier lifestyle and preserving their kidney function.

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